

Name:

Solutions

Date:



SAFETY BAY
SENIOR HIGH SCHOOL
imagine believe achieve

Year 9 Mathematics

Test 3, 2015

Topic – Surface Area, Volume and Capacity

30

= _____ %

Total Time 60 minutes

Weighting: 10% of the year.

Equipment:

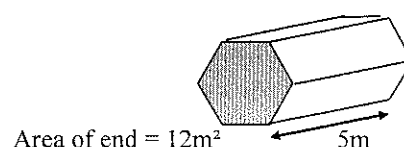
Page notes (A4 one side),
Scientific Calculator

1. (2 Mark)

Calculate the Volume of the Prism

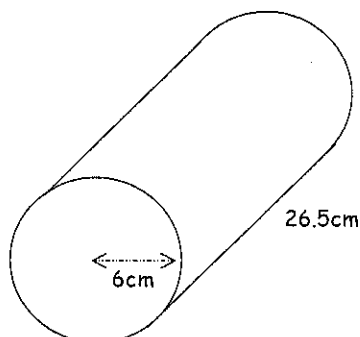
$$12 \times 5 = 60 \text{ m}^3 \quad \checkmark \text{ (answer)}$$

$$\checkmark \text{ (units)}$$



2. (5 Marks: 2, 2, 1)

Use the cylinder below to answer the following questions



a. Calculate the Cylinders Surface Area
(to nearest cm^2)

$$\begin{aligned} \text{S.A.} &= 2\pi r^2 + 2\pi r h \\ &= 2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times 26.5 \\ &= 1225 \text{ cm}^2 \end{aligned} \quad \checkmark \checkmark$$

b. Calculate the Volume
(to nearest cm^3)

$$\begin{aligned} V &= 2\pi r^2 h \\ &= 2 \times \pi \times 6^2 \times 26.5 \\ &= 5994 \text{ cm}^3 \end{aligned} \quad \checkmark \checkmark$$

c. What is the Capacity of the Cylinder in mL's?

$$5994 \text{ mL} \quad \checkmark$$

3. (4 Marks)

Calculate the Volume of the Triangular Prism

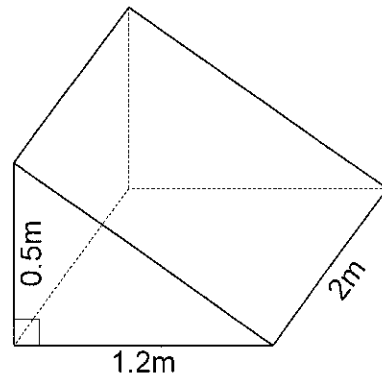
$$V = \frac{1}{2} b h \times \text{length}$$

$$= \frac{1}{2} \times 1.2 \times 0.5 \times 2$$

$$= 0.6 \text{ m}^3$$

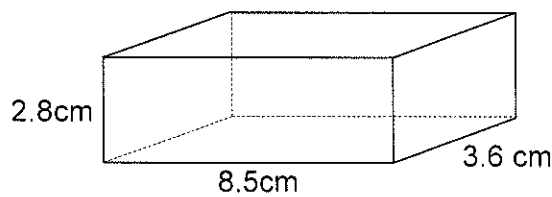
✓✓ (answer)

✓(units)



4. (4 Marks)

What is the total Surface Area of this box?

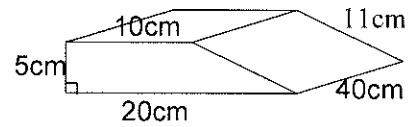


$$\begin{array}{lcl}
 \text{Back/Front} & = & 2.8 \times 8.5 \times 2 = 47.6 \\
 \text{Left/Right} & = & 2.8 \times 3.6 \times 2 = 20.16 \\
 \text{Top/Bottom} & = & 8.5 \times 3.6 \times 2 = 61.20
 \end{array}
 \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \\ \\ \end{array} \text{✓✓ (-1 per error)}$$

$$\begin{array}{r}
 47.6 \\
 20.16 \\
 \hline
 61.20 \\
 \hline
 128.96 \text{ cm}^2 \text{ ✓✓}
 \end{array}$$

5. (5 Marks)

Calculate the Surface Area of the following Trapeziod.



$$2 \times \left(\frac{10+20}{2} \right) \times 5 = 150 \checkmark$$

$$\text{Top} = 10 \times 40 = 400$$

$$\text{Back} = 5 \times 40 = 200$$

$$\text{Bottom} = 20 \times 40 = 800$$

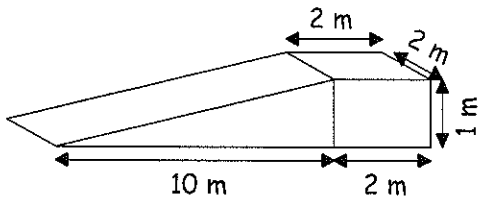
$$\text{slant} = 11 \times 40 = 440$$

$$\text{Total Surface Area} = 1990 \text{ cm}^2 \checkmark \checkmark$$

$\checkmark \checkmark$ (-1 per error)

6. (4 Marks)

A ramp is constructed as shown. Calculate the volume of concrete needed to build the ramp.



$$\text{Trapezium} = \left(\frac{2+12}{2} \right) \times 1$$

$$= 7 \text{ m}^2 \checkmark$$

$$\text{Vol} = 7 \times 2$$

$$= 14 \text{ m}^3 \checkmark \checkmark$$

$\checkmark \checkmark$ or similar

They will require 14 m^3 of concrete $\checkmark$

7. (6 Marks: 1, 5)

A winery stores its wine in wooden barrels that are 80 cm in diameter and 120cm tall.

- a. What geometric shape could the barrels be assumed to have?

Cylinder ✓

- b. How many 750mL wine bottles could be filled from a full barrel? Show all working for full marks.

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi \times 40^2 \times 120 \\ &= 603185.7895 \text{ cm}^3 \\ &\text{or } 192000\pi \end{aligned} \quad \checkmark \checkmark$$

$$\text{Capacity of Barrel} = 603185.8 \text{ ml} \quad \checkmark$$

$$\begin{aligned} \text{Number of wine bottles} &= \frac{603185.8}{750} \\ &= 804.24 \quad \checkmark \\ &= 804 \text{ bottles} \end{aligned}$$

They can fill 804 wine bottles from the barrel. ✓

End of Test